

DIPLOMA PLC

mycarbon.co.uk

METHODOLOGY MANUAL



Executive Summary

Reporting Period: Fiscal Year 2025 (FY25)

Standard Alignment: GHG Protocol Corporate Standard

This document outlines the calculation methodologies, assumptions, and data sources utilised to quantify the Carbon Footprint for Diploma PLC for the Fiscal Year 2025. The approach detailed herein is designed to ensure accuracy, transparency, and consistency, enabling a robust year-on-year comparison against historical baselines.

The methodologies applied in FY25 adhere to the principles of the GHG Protocol Corporate Standard and ISO 14064-1, prioritising the use of primary activity data (e.g., kWh, weight, distance) where available. In instances where primary data was inaccessible, spend-based estimates or rigorous forward-extrapolations based on audited revenue figures were applied to ensure reporting completeness.

Key Reporting Principles

- **Consistency:** The core methodology remains consistent with previous reporting years to maintain the integrity of historical trend analysis.
- **Accuracy:** Significant refinements have been made this year, specifically regarding End-of-Life treatment for sold products, incorporating updated material composition and recycling rates to reflect modern industrial waste practices.
- **Completeness:** This methodology covers Scope 1 and 2 emissions alongside Scope 3 categories including Purchased Goods, Capital Goods, Logistics, Business Operations, and Product Lifecycle.
- **Transparency:** All data limitations, specifically regarding late reporting from entities such as Windy City Wire and Somagen, have been explicitly documented with their corresponding extrapolation logic (see [5. Data Limitation & Extrapolation Summary](#)).

Statement of Validity This methodology manual correctly reflects the operational boundaries, emission factors, and calculation logic applied to the FY25 Carbon Footprint. It serves as the primary reference for internal stakeholders and external auditors regarding the quantification of the FY25 inventory.

Contents

EXECUTIVE SUMMARY.....	0
1. SCOPE 1 & 2: DIRECT INDIRECT EMISSIONS.....	3
SCOPE 1: DIRECT EMISSIONS	3
SCOPE 2: ENERGY INDIRECT EMISSIONS	3
<i>Electricity – Location-Based Method</i>	3
<i>Electricity – Market-Based Method</i>	4
<i>District Heating</i>	4
<i>Electric Vehicles (Scope 2)</i>	4
2. SUPPLY CHAIN & LOGISTICS.....	5
PURCHASED GOODS & SERVICES.....	5
CAPITAL GOODS.....	5
UPSTREAM TRANSPORTATION & DISTRIBUTION.....	5
3. BUSINESS OPERATIONS.....	6
WASTE GENERATED IN OPERATIONS	6
BUSINESS TRAVEL	6
EMPLOYEE COMMUTING	6
4. PRODUCT LIFECYCLE (DOWNSTREAM)	7
END OF LIFE OF SOLD PRODUCTS.....	7
5. DATA LIMITATION & EXTRAPOLATION SUMMARY ERROR! BOOKMARK NOT DEFINED.	

1. Scope 1 & 2: Direct Indirect Emissions

Scope 1: Direct Emissions

Scope 1 accounts for all direct greenhouse gas emissions arising from sources owned or controlled by the reporting entity.

Methodology: Activity-based (recalculation).

Approach: Data is collected per entity code and specific operating location. To ensure auditability, MyCarbon performs a complete recalculation of all client-provided figures.

- **Data Source:** Primary activity data (fuel consumption, distance, refrigerant logs) provided per entity code and operating location.
- **Calculation Logic:** Emissions tCO₂e = Activity Data x Emission Factor.
- **Unit Conversions:** Standard unit conversions (e.g., gallons to litres, miles to km) are applied prior to calculation using DEFRA conversion tables.

Category Specifics:

- **Stationary & Mobile Combustion:** Covers natural gas, LPG, fuel oil, and vehicle fuels.
- **Electric Vehicles (EVs):** Zero Scope 1 emissions are attributed to EVs (no tailpipe emissions). Charging electricity is accounted for in Scope 2.
- **Refrigerants:** Calculated based on the mass of gas lost or replenished during the reporting year. Global Warming Potentials (GWPs) are applied as per DEFRA FY25.

Scope 2: Energy Indirect Emissions

Scope 2 covers indirect emissions associated with the generation of purchased electricity and district heating. Calculations are dual-reported using both Location-Based and Market-Based methods in alignment with the GHG Protocol.

Methodology: Dual-Reporting (Location-Based & Market-Based).

Approach: As with Scope 1, data is provided per entity code and location, then independently recalculated by MyCarbon. Raw consumption figures (kWh) and evidence of green procurement (RECs/REGOs).

Electricity – Location-Based Method

Reflects the average grid mix intensity in the area of consumption.

- **Non-US Sites:** IEA location-based emission factors.
- **US Sites:** EPA (eGRID) emission factors applied at the relevant regional grid level.

Electricity – Market-Based Method

Reflects the emissions from electricity that Diploma PLC has purposefully chosen (contractual instruments). The calculation follows this hierarchy:

1. **Renewable Certificates:** Entities with verified renewable electricity (RECs, REGOs, or equivalent) are assigned a factor of 0 kgCO₂e/kWh.
2. **Residual Mix:** For sites without certificates, a residual mix factor is applied:
 - *Europe:* AIB residual factors.
 - *USA:* Green-e residual factors.
3. **Fallback:** If a residual mix factor is unavailable for a specific region, the Location-Based factor is applied.

District Heating

Emissions are calculated using the best available proxy for the local or national district heating supply mix.

- **Prioritisation:** Emission factors are selected from DEFRA, EEA, or UBA (Germany) based on geographical relevance to the point of consumption.

Electric Vehicles (Scope 2)

Electricity consumed for EV charging is calculated under Scope 2 using DEFRA 2025 distance-based emission factors where direct metering is unavailable.

2. Supply Chain & Logistics

Purchased Goods & Services

Methodology: Spend-based.

Approach: The calculation is grounded in an analysis of the purchase ledger. To ensure accuracy, deliberate exclusions were applied to filter out financial entries that do not correspond to GHG emissions.

- **Exclusions:** Intangible purchases, account adjustments, depreciation, stock provisions, and non-consumptive financial entries.
- **Inclusions:** Payments to contractors and temporary staff have been included this year in limited instances (a refinement from previous years).

Capital Goods

Methodology: Spend-based.

Approach: Emissions are estimated directly from capital expenditure (CapEx) figures. This approach remains consistent with previous assessments.

- **Process:** Relevant CapEx data is examined, and appropriate emission factors are assigned to quantify the carbon impact of these investments.

Upstream Transportation & Distribution

Methodology: Hybrid (Activity-based & Spend-based).

- **Standard Approach:** Data from expenditure records determines emissions. Detailed activity data is prioritised where available.
- **Specific Entity Treatments:**
 - **Windy City Wire (WCW) & Dicsa Spain:** Recorded expenditure is netted out where detailed activity data exists.
 - **General Ledger Allocations:** Where direct activity data is absent, specific ledger tabs ("Freight In" / "Freight Out") are utilised.
- **Mode Assumptions:** A breakdown of 65% Air and 35% Road is applied to general freight spend. DEFRA factors for rail, land, water, and air are utilised.
- **Data Limitation & Extrapolation (WCW):**

Note: Logistics data from UPS for Windy City Wire was unavailable at the time of reporting.

Mitigation: A forward extrapolation was applied based on FY24 activity data, adjusted for WCW-specific revenue growth.

3. Business Operations

Waste Generated in Operations

Methodology: Activity-based (Primary) / Spend-based (Secondary).

Approach:

- **Primary:** Derived from the Diploma activity database (quantitative weights). Standard emission factors applied.
- **Secondary:** Where weight data was unavailable, spend-based emission factors were utilised.

Business Travel

Methodology: Spend-based.

Approach: Consistent with previous years, travel emissions are calculated by analysing detailed expenditure reports.

- **Categorisation:** Spend is broken down into Air, Rail, and Other Modes.
- **Calculation:** Corresponding emission factors are applied to each financial category to estimate the footprint.

Employee Commuting

Methodology: Survey-based (FY24 Extrapolation).

Data Source: FY24 Employee Survey results applied to FY25 headcount.

Calculation Logic: The per-employee emission intensity (derived from FY24) was applied to the current staff numbers. The FY24 survey followed this specific protocol:

1. **Distance:** One-way distance x2 (Round trip).
2. **Frequency:** Multiplied by number of commuting days per week.
3. **Annualization:** Multiplied by 49 working weeks.
4. **Adjustments:**
 - **Outlier Removal:** Single commuting distances >150 km excluded.
 - **Data Cleaning:** Non-commuting responses with contradictory data removed.
 - **Scaling:** Results adjusted based on the response ratio per site to reflect total staff numbers.
 - **Homeworking:** Included using a similar calculation logic.

4. Product Lifecycle (Downstream)

Use of Sold Products

Methodology: Activity-based (Wattage x Usage).

Data Source: "Machine Sold" file (Unique product/serial codes with wattage).

Calculation Formula: Total kWh = Wattage x 7 hours/day x 365 days/year} x 10 years (Lifespan)

Regional Variations:

- **Australia/New Zealand:** Updated report reflects a shift in power consumption and sales patterns (move from rental/placement to exclusive sales).
- **Canada (Somagen):** FY25 report unavailable.

Mitigation: FY24 consumption figures were extrapolated forward based on Somagen revenue growth (FY25 vs FY24).

End of Life of Sold Products

Methodology: Material Composition & Intensity Modelling.

Regional Approaches:

- **Australia:**
 - **Input:** Individual product specifications (Weight).
 - **Material Assumption:** 50% Metal / 50% Plastic.
 - **Disposal Assumption:** 80% Recycling / 20% Landfill.
 - **Context:** This is a methodology update (previously 100% landfill) to reflect the high material value of industrial machines and modern regulation.
- **Canada:**
 - **Input:** Unavailable.
 - **Mitigation:** FY24 results extrapolated based on Somagen revenue growth.
- **Other Entities:**
 - **Input:** Unavailable.
 - **Estimation Method:** A "Weight of Sold Products Intensity" was calculated using different businesses outbound weights vs. outbound logistics spend.

- This intensity metric was applied to the logistics spend of relevant entities to estimate total weight.
- The standard assumptions (50/50 mix, 80/20 recycling/landfill) were then applied.